

NOTES FOR TALK ON SHEARING

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This talk will compare different logical notions of when a formula imposes a constraint which is “non-generic” or “too small to be visible” to a certain set of parameters, dividing and shearing. The first of these notions specializes to familiar notions of independence / genericity in familiar mathematical structures such as purely algebraically closed fields, and can be used to carry such ideas to other structures, such as difference / differential fields. The behavior of dividing and shearing in a given theory also tells us something about the combinatorics of definable relations in a given theory, and gives us tools for comparing theories to each other.

1. DIVIDING AND SIMPLICITY

1.1. Dividing and Forking. We begin with dividing, and the setting where dividing is most useful: simple theories.

Definition. A formula $\varphi(x; a)$ *divides* over a parameter set B if there is an B -indiscernible sequence $(a_i)_{i < \omega}$ with $a = a_0$ such that $\{\varphi(x; a_i) \mid i < \omega\}$ is inconsistent.

Here a sequence $(a_i)_{i < \omega}$ is A -indiscernible if for any $i_1 < \dots < i_n$ and $j_1 < \dots < j_n$, we have

$$(a_{i_1}, \dots, a_{i_n}) \equiv_A (a_{j_1}, \dots, a_{j_n})$$

meaning that these tuples satisfy the same formulas with parameters from A .

Example 1. Suppose we are working the field $\mathbb{C}$, $A = \mathbb{Q}$, and $\varphi((x, y); b, c)$ is the formula $y = bx + c$, defining a line in the plane, where b and c are independent transcendentals. Choose (b_i, c_i) so that

$$\{b_i \mid i < \omega\} \cup \{c_i \mid i < \omega\}$$

is an algebraically independent set. Then $(b_i, c_i)_{i < \omega}$ is $\mathbb{Q}$ -indiscernible in $\mathbb{C}$. For any pair (b_i, c_i) (b_j, c_j) the lines $y = b_i x + c_i$ and $y = b_j x + c_j$ intersect, but any triple have empty intersection. This shows that the formula divides over $\mathbb{Q}$.

What we saw here is a typical example of dividing.

Proposition. *The following are equivalent:*

- (1) $\varphi(x; b)$ *divides* over A (there is a fully indiscernible sequence along which we have inconsistency).
- (2) There is a sequence $(b_i)_{i < \omega}$ such that $b_i \equiv_A b$ for each i , and

$$\{\varphi(x; b_i) \mid i < \omega\}$$

is k -inconsistent for some k : for any $i_1 < \dots < i_k$, $\varphi(x; a_{i_1}) \wedge \dots \wedge \varphi(x; a_{i_k})$ is inconsistent. (There is a 1-indiscernible sequence along which we have highly uniform inconsistency, that is k -inconsistency.)

To get from (2) to (1), we use Ramsey’s theorem to find an indiscernible sequence which copies certain properties of the sequence $(a_i)_{i < \omega}$ which already hold uniformly, including k -inconsistency. We will return to this point later in the talk.

If we think of the sets defined by dividing formulas as “small,” then it makes sense to also consider finite unions of such sets small.

Definition. A formula φ *forks* over B if $\varphi \vdash \psi_1 \vee \dots \vee \psi_n$, where each ψ_i divides over B .

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We say that a type p forks/divides over B if it implies a formula which forks/divides over B . In complete generality, forking enjoys the following advantage over dividing.

Proposition. *If $p \in S(B)$ does not fork over A , then for any $C \supset B$, there is a $q \in S(C)$ extending p such that q does not fork over A .*

Using forking, we can define a notion of when two tuples are independent relative to a set of parameters:

$$a \downarrow_C b \iff \text{tp}(a/Cb) \text{ does not fork over } C$$

Intuitively, this should mean something like “ Cb does not have any essentially new information about a that C does not already have.” How well this intuition accords with reality depends on properties of the theory.

Remark. For sets A, B , we say $A \downarrow_C B$ if $a \downarrow_C b$ for any tuples $a \subset A$, $b \subset B$.

1.2. Simple Theories. In the example of dividing we say above in a field, there was an associated drop in dimension. Without φ , all we knew about (x, y) is that it was a point in the plane, which has dimension 2. The dividing formula asserts that (x, y) lies on a certain line, which has dimension 1. In fact, in algebraically closed fields, dividing/forking always corresponds to a drop in dimension. This tells us, in particular, that we cannot have an infinite sequence of repeated dividing. The property of simplicity is essentially this, at least applied to each formula at a time.

Definition. A theory T is said to be *simple* if there is no formula $\varphi(x; y)$, sequence $(a_i)_{i < \omega}$, and integer k such that:

- (1) $\{\varphi(x; a_i) \mid i < \omega\}$ is consistent.
- (2) $\varphi(x; a_i)$ k -divides over $\{a_j \mid j < i\}$ for each i .

There are many equivalent characterizations of simplicity:

Proposition. *The following are equivalent, for T a complete theory:*

- (1) T is simple.
- (2) *Dividing has a local character in T :* for any type $p \in S(B)$, there is an $A \subset B$ of size $|A| \leq |T|$ such that p does not divide over A .
- (3) *No formula $\varphi(x; y)$ has the tree property, meaning that there is a k and parameters $(a_\eta)_{\eta \in \omega^{<\omega}}$ such that*
 - (a) *Paths are consistent:* for $\pi \in \omega^\omega$, $\{\varphi(x; a_{\pi \upharpoonright n}) \mid n < \omega\}$ is consistent.
 - (b) *Siblings are k -inconsistent:* for each η , $\{\varphi(x; a_{\eta \smallfrown i}) \mid i < \omega\}$ is k -inconsistent.

Example 2. Torsion free divisible abelian groups ($\mathbb{Q}$ -vector spaces) $(V, +, 0)$ are simple.

$$A \downarrow_C B \iff \text{span}(AC) \cap \text{span}(BC) = \text{span}(C)$$

Example 3. Algebraically closed fields $(k, +, \cdot, 0, 1)$ are simple.

$$A \downarrow_C B \iff \text{for any } a_1, \dots, a_n \in A, \text{tr.deg}(a_1, \dots, a_n/\mathbb{Q}(BC)) = \text{tr.deg}(a_1, \dots, a_n/\mathbb{Q}(C))$$

Example 4. The random graph (G, E) is simple.

$$A \downarrow_C B \iff A \cap B \subset C$$

Example 5. The dense linear order $(\mathbb{Q}, <)$ is not simple. Indeed, the formula $y_1 < x < y_2$ has the tree property (draw picture).

1.3. The Kim-Pillay Theorem. In simple theories, dividing and forking satisfy certain properties that accord well with the intuitive description I gave of them, and generalize important properties of the special cases we saw above. First of all, we have:

Proposition (Kim). *In simple theories, forking is equivalent to dividing.*

Moreover, we have the following structural properties of $\downarrow$:

(1) Monotonicity: if $A \downarrow_C B$, and $A' \subset A$, $B' \subset B$, and $C \subset C' \subset C \cup A \cup B$, then $A' \downarrow_{C'} B'$.

(2) Symmetry: $a \downarrow_C b \iff b \downarrow_C a$.

(3) (Right) Transitivity: for $B_1 \subset B_2 \subset B_3$, we have

$$a \downarrow_{B_1} B_3 \iff a \downarrow_{B_1} B_2 \text{ and } a \downarrow_{B_2} B_3$$

(4) (Left) Transitivity:

$$a_1 a_2 \downarrow_C B \iff a_1 \downarrow_C B \text{ and } a_2 \downarrow_{Ca_1} Ba_1$$

Note that this is equivalent to right transitivity if symmetry holds, so is redundant here.

The existence of nonforking extensions and local character of dividing/forking can also be phrased as structural properties of $\downarrow$, as can certain finiteness properties, which trace back to dividing/forking being witnessed by a finite formula.

These properties, together with one other, characterize forking in simple theories in the following sense.

Theorem 1 (Kim-Pillay 1997). *If in a theory T , we have an abstract independence relation which satisfies the structural properties mentioned above, as well as an additional property: the Independence Theorem over Models, then T is simple and the relation is nonforking independence.*

2. RAMSEY CLASSES AND GENERALIZED INDISCERNIBLES

In their study of the interpretability order, a certain partial order on theories, Malliaris and Shelah considered models built up from generalized indiscernibles, and inconsistency of formulas along such indiscernibles.

Definition. Let I be an (ordered) index structure, for the purposes of this talk a relational structure. An indexed family $(a_t)_{t \in I}$ is said to be *generalized indiscernible* over B if for any $t_1 < \dots < t_n$, $s_1 < \dots < s_n$ from I ,

$$(t_1, \dots, t_n) \simeq (s_1, \dots, s_n) \implies (a_{t_1}, \dots, a_{t_n}) \equiv_B (a_{s_1}, \dots, a_{s_n})$$

This is to say, the type over B of a tuple $(a_{t_1}, \dots, a_{t_n})$ depends only on the induced structure on $\{t_1, \dots, t_n\}$.

When working with sequences, Ramsey's theorem allows us to find indiscernible sequences copying various properties of sequences we already have. To be able to find generalized indiscernibles in a similar way, we need an analogue of Ramsey's theorem to hold.

Definition. A class K of finite structures (satisfying the hereditary and joint embedding properties) is called a *Ramsey class* if for any A and B in K and integer r there is a C in K such that given a coloring

$$f : \binom{C}{A} \rightarrow r$$

of copies of A in C with r colors, there is a copy $\tilde{B} \subset C$ of B such that f is constant on the set of copies of A in $\tilde{B}$.

Example 6. Ramsey's theorem states that the class of finite linear orders is a Ramsey class.

Theorem 2 (Nešetřil-Rödl). *Fix r and n .*

- (1) *The class of ordered r -hypergraphs is a Ramsey class.*
- (2) *The class of ordered r -hypergraphs omitting an n -clique is a Ramsey class.*

Remark. What Nešetřil and Rödl prove is actually more general than this. There are also many constructions that take existing Ramsey classes and produce new ones. Together with the Nešetřil-Rödl theorem, these can be used to produce many examples of Ramsey classes.

For an index structure I , $\text{age}(I)$ is the class of finite substructures of I . Given a family $(a_t)_{t \in I}$ indexed by I , we can for any J with $\text{age}(J) = \text{age}(I)$ find a corresponding family $(a_t)_{t \in J}$ which copies all of its uniform properties, so we consider the class $\text{age}(I)$. We will be able to find lots of indiscernibles indexed by I provided that $\text{age}(I)$ is Ramsey.

Remark. With a slightly different definition of shearing, the Ramsey assumption can be dropped from many of the theorems which follow. I will keep the Ramsey assumption for this talk, as this is needed to relate shearing to generalized indiscernibles, the original context in which it was introduced.

3. SHEARING

3.1. First Definition. Let K be a Ramsey class, $\varphi(x; a)$ a formula and B a parameter set.

Definition. We say that $\varphi(x; a)$ K -shears over B if there is an I with $\text{age}(I) = K$, a generalized indiscernible $(a_t)_{t \in I}$ over B , where $a = a_{t_0}$ for some $t_0 \in I$, and

$$\{\varphi(x; a_t) \mid t \in I\}$$

is inconsistent.

Remark. We will assume henceforth that our Ramsey class does not have distinguishable singleton elements. If it does, we could arrange so that $\varphi(x; a_s)$ is inconsistent for s distinguished from t_0 . This would give us lots of shearing essentially for free, which tells us nothing about $\varphi(x, a)$.

Remark. This definition of shearing is not the one given by Malliaris and Shelah. We will address the discrepancy between the definitions soon.

As an example, consider the Henson graph, the Fraïssé limit of triangle free graphs. The formula $\varphi(x; y) = xEy$ exhibits the following instance of shearing with respect to K , the class of ordered triangle free graphs. By quantifier elimination any copy H of the Henson graph can be viewed as a generalized indiscernible indexed by putting some ordering on it. The formulas xEa and xEb are inconsistent whenever a is connected to b in H . This is an example of shearing which does not come from dividing in the Henson graph. If we restrict our attention to simple theories, nothing like this can happen:

Proposition. *If T is a simple theory, then for K either the class of ordered r -hypergraphs or ordered r -hypergraphs omitting a clique, K -shearing is equivalent to dividing in T .*

We will need to find new Ramsey classes before we can see nontrivial examples within simple theories. First, though, we can learn something from this example. As with dividing, the indiscernibility of the parameters yielding inconsistent instances of our formula implies that the inconsistency appears with a high level of uniformity.

Definition. An indexed family $(\varphi_t)_{t \in I}$ is said to be F -inconsistent for $F \in \text{age}(I)$ if $\varphi_{t_1} \wedge \cdots \wedge \varphi_{t_n}$ is inconsistent whenever $\{t_1, \dots, t_n\}$ is a copy of F .

Proposition. *Let K be a Ramsey class, $\varphi(x; a)$ a formula, and B a set.*

- (1) *If $\varphi(x; a)$ K -shears over B , this can be witnessed using any I with $\text{age}(I) = K$.*
- (2) *$\varphi(x; a)$ K -shears over B if and only if there is an indexed family $(a_t)_{t \in I}$ such that*
 - (a) *$a_t \equiv_B a$ for each $t \in I$.*
 - (b) *$(\varphi(x; a_t))_{t \in I}$ is F -inconsistent for some F .*

Because F -inconsistency is a weaker condition than k -inconsistency, we see that dividing implies shearing. We can view shearing as obtained by relaxing the uniformity with which inconsistency appears in dividing. When the inconsistency witnessing K -shearing is F -inconsistency, we say that the formula (K, F) -shears.

3.2. New Ramsey Classes and Second Definition. To see nontrivial shearing in simple theories, we need to go back to the original definition of shearing.

Definition (Malliaris-Shelah). A *context* is a pair $(\mathcal{K}, I)$, where $\mathcal{K}$ is a class of structures whose collective age is a Ramsey class, and $I \in \mathcal{K}$ is a structure. Given finite structures $F \subset G \subset I$, a formula $\varphi(x; a)$ is

said to (F, G) -shear with respect to the context over B if there is a $J \supset I$ which is ω -homogeneous and ω -universal within $\mathcal{K}$, and a generalized indiscernible

$$(a_{G'} : F \subset G' \subset I \text{ and } (F, G) \simeq (F, G'))$$

over B , such that $a = a_G$ and such that $\{\varphi(x; a_{G'}) \mid G' \dots\}$ is inconsistent.

In other words, we allow our generalized indiscernible to be indexed by structures varying over a fixed base, rather than individual elements. To bridge the gap between these definitions, we need to introduce new Ramsey classes whose elements parameterize such structures in a given Ramsey class.

Definition. Let K be a Ramsey class, and $F \subset G$ structures in K . For $H \in K$ and a choice of copy of F , $F_0 \subset H$, $H[G/F_0]$ is the set of copies of G extending this fixed copy of F . We make this a relational structure by recording all the relations between all the elements of these various copies of G . $K[G/F]$ is the class of substructures of $H[G/F_0]$, as H ranges over K .

Proposition. *If K is a Ramsey class, $K[G/F]$ is a Ramsey class. Shearing in the sense of Malliaris-Shelah can be viewed as shearing with respect to $K[G/F]$.*

With this construction, we can see some examples which are not dividing.

3.3. Examples in Simple Theories.

Remark. In the notation $K[G/F]$, if F is the empty structure (which it will be in all these examples,) we omit it.

Let L be the class of linear orders. Then L -shearing is exactly dividing. Letting 2 denote a 2-element linear order, we have a new Ramsey class $L[2]$. Structures in this consist of pairs of elements of a linear order, and we keep track of the relative order of the various elements of various pairs.

In the random graph, consider the formula $\varphi(x; y, z) = xEy \wedge x \not E z$. Take an infinite anticlique $(v_i)_{i < \omega}$. Then $\omega[2] = \{(i, j) \mid i < j < \omega\}$ has age $L[2]$, and $a_{(i, j)} = (v_i, v_j)$ is generalized indiscernible over $\emptyset$. The formulas $\{\varphi(x; a_{(i, j)}) \mid (i, j) \in \omega[2]\}$ are inconsistent, and in fact we have inconsistency in any pair of the form $(i, j), (j, k)$. All dividing in the random graph comes from equations between variables, so this is an instance of shearing which is not dividing. Essentially all forms of nontrivial shearing in the random graph look like this example.

There is a somewhat more complex Ramsey class K such that the generic tetrahedron free 3-hypergraph, the Fraïssé limit of finite tetrahedron free 3-hypergraphs has nontrivial K -shearing. This Ramsey class can be chosen so that the random graph has no nontrivial K -shearing.

In detail:

Definition. Let K be the class of ordered, 3-colored, tetrahedron free 3-hypergraphs where all red vertices precede all blue vertices precede all yellow vertices. Let $H \in K$ be a triple $\{r, b, y\}$, with no edge connecting them. The Ramsey class we will work with is $K[H]$.

Consider the formula

$$\varphi(x; u, v, w) = R(x, u, v) \wedge R(x, u, w) \wedge R(x, v, w)$$

Then for vertices a, b, c not connected by an edge, $\varphi(x; a, b, c)$ $K[H]$ -shears but does not divide over $\emptyset$. We index triples (a', b', c') by copies of H in a straightforward way, and we get inconsistency whenever we have the following configuration:

[Draw picture.]

Remark. These examples appear, with a somewhat different presentation, in the second paper of Malliaris and Shelah on shearing.

4. A DICHOTOMY FOR SHEARING

So far, we have seen that many forms of shearing are equivalent to dividing, but that other forms of shearing can occur even in theories where dividing is trivial. In the latter case, we can ask how much shearing is like dividing, in particular, whether it can be used to define a stronger independence relation sharing some of the properties of nonforking independence.

Question. Which of the Kim-Pillay axioms are satisfied by shearing in simple theories? We could ask the same question about the analogue of forking.

First of all, some axioms will be true, falling out of the form of the definition of shearing with no special considerations. In particular, the monotonicity and finite character axioms are true. Less immediately, the left transitivity axiom is true: if $\text{tp}(a_1/B)$ does not K -shear over C , and $\text{tp}(a_2/Ba_1)$ does not K -shear over Ca_1 , then $\text{tp}(a_1a_2/B)$ does not shear over C .

We have also seen that K -shearing for many Ramsey classes K is just equivalent to dividing in simple theories, and so will satisfy the Kim-Pillay axioms. It turns out that in all other cases, shearing will fail to satisfy analogues of local character, right transitivity and symmetry. To make this precise, we state here what these analogues are.

Definition. We say that K -shearing satisfies symmetry in T if given a, b and a parameter set C , $\text{tp}(a/Cb)$ does not K -shear over C if and only if $\text{tp}(b/Ca)$ does not K -shear over C . Likewise, we say that the ideal generated by K -shearing formulas (the analogue of forking) satisfies symmetry if $\text{tp}(a/Cb)$ does not imply a finite disjunction of K -shearing formulas if and only if $\text{tp}(b/Ca)$ does not.

Definition. We say that K -shearing satisfies right transitivity in T if given $a, B_1 \subset B_2 \subset B_3$, if $\text{tp}(a/B_2)$ does not K -shear over B_1 and $\text{tp}(a/B_3)$ does not K -shear over B_2 , then $\text{tp}(a/B_3)$ does not K -shear over B_1 . We could likewise define right transitivity for the ideal generated by K -shearing formulas.

Proposition. *If K -shearing is not equivalent to dividing in T , there are counterexamples to all of these properties, as well as to an analogue of the local character property.*

This falls out of a more general dichotomy concerning the behavior of shearing. We first formulate the analogue of simplicity.

Proposition. *The following are equivalent:*

- (1) *There is no formula φ , $F \in K$ and sequence $(a_i)_{i < \omega}$ such that $\{\varphi(x; a_i) \mid i < \omega\}$ is consistent and each $\varphi(x; a_i)$ (K, F) -shears over $\{a_j \mid j < i\}$.*
- (2) *No formula satisfies an analogue of the tree property patterned on the tree $I^{<\omega}$ with $\text{age}(I) = K$, and with F -inconsistency replacing k -inconsistency.*
- (3) *K -shearing has a local character: any type does not K -shear over a parameter set of bounded size.*

We call theories with these properties K -simple.

The dichotomy for shearing is summarized in the following theorem.

Theorem 3. *The following are equivalent:*

- (1) *T is K -simple.*
- (2) *T is simple and K -shearing is equivalent to dividing in T .*
- (3) *K -shearing, or the ideal generated by K -shearing formulas, satisfies either symmetry, right transitivity, or local character in T .*

The property of K -simplicity always implies simplicity. On the other hand, the stronger property of stability of T implies K -simplicity for every K : the tree witnessing a failure of K -simplicity also witnesses a failure of stability. Thus in stable theories, all forms of shearing are equivalent to dividing.

Example 7. For L the class of finite linear orders, L -shearing is dividing and L -simplicity is simplicity. For K_r the class of ordered r -hypergraphs, and $K_{r,n}$ the class of ordered n -clique omitting r -hypergraphs, K_n -simplicity and $K_{n,r}$ -simplicity are also equivalent to simplicity.

Example 8. $L[2]$ -simplicity is equivalent to stability.

Example 9. For K the Ramsey class having nontrivial shearing in the generic tetrahedron free 3-hypergraph, K -simplicity lies between stability and simplicity. The random graph is an unstable theory which is K -simple, but the generic tetrahedron free 3-hypergraph is not.

5. QUESTIONS FOR FUTURE WORK

One question unanswered by the above is when K -shearing satisfies the extension axiom in the Kim-Pillay theorem. This is related to the question of whether K -shearing formulas form an ideal, analogous to the question of whether forking is equivalent to dividing. By the way, for the independence theorem, I have an example of it failing, but no general theorem.

What we have discussed establishes many equivalent characterizations of the property of K -simplicity, for K a Ramsey class. I would like to understand better when this property does or does not hold in specific simple theories. A particularly interesting case is that of pseudofinite fields.

A pseudofinite field is a field k such that for any sentence φ in the language of rings with $k \models \varphi$, there is a finite field F_q such that $F_q \models \varphi$. Ax gave an algebraic characterization of these fields as those which are perfect, pseudo-algebraically closed, and which have a unique cyclic extension of degree n for each n .

Theorem 4 (Hrushovski). *Any pseudofinite field is simple. The independence relation is the same as in algebraically closed fields.*

Theorem 5 (Beyarslan). *For any r , the generic r -hypergraph is interpretable in any pseudofinite field.*

Theorem 6 (Hrushovski). *The generic tetrahedron free 3-hypergraph is not interpretable in any pseudofinite field.*

Given that shearing can be used to draw a distinction between these combinatorial theories, these results suggest the following question.

Question. What kinds of nontrivial shearing can occur in pseudofinite fields? In other words, for which K are pseudofinite fields K -simple?

Proposition. *Suppose that in a pseudofinite field, a formula (K, F) -shears but does not divide over some parameter set. Consider the class of hypergraphs K_F , which consists of hypergraphs of the form G_F obtained by placing an edge on every copy of F , for $G \in K$. This class has linear size anticliques, in the sense that there is a fixed $\lambda > 0$ such that for any G , G_F has an anticlique of size at least $\lambda|G|$.*

Remark. This is a strong condition, because it applies to any hypergraph in a hereditary class of hypergraphs. It is stronger than omitting cliques of a given size, for example, triangle free graphs of size n are only guaranteed to have anticliques of size about $\sqrt{n}$.

Question. Is this enough to recover Hrushovski's theorem?